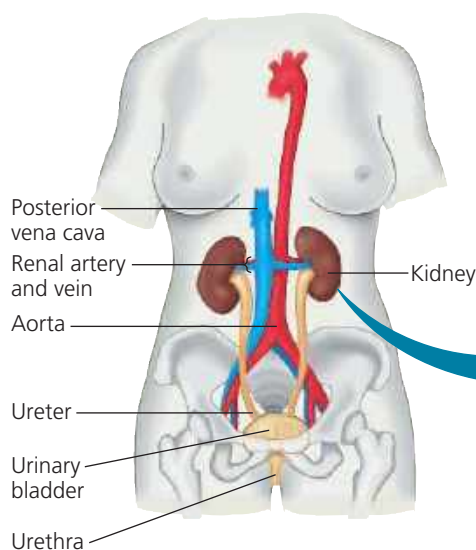
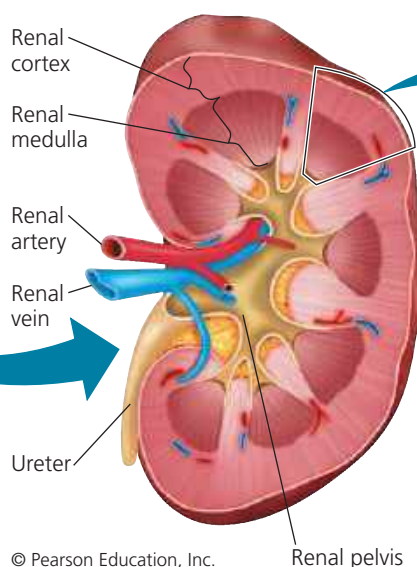


▼ Figure 44.12 Exploring the Mammalian Excretory System

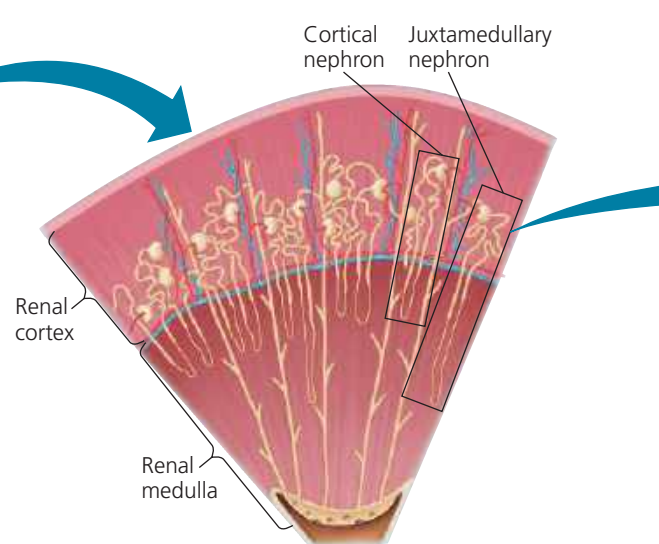
Excretory Organs



Kidney Structure



Nephron Types



In humans, the excretory system consists of **kidneys**, a pair of organs each about 10 cm in length, as well as organs for transporting and storing urine. Urine produced by each kidney exits through a duct called the **ureter**; the two ureters drain into a common sac called the **urinary bladder**. During urination, urine is expelled from the bladder through a tube called the **urethra**, which empties to the outside near the vagina in females and through the penis in males. Sphincter muscles near the junction of the urethra and bladder regulate urination.

Each kidney has an outer **renal cortex** and an inner **renal medulla**. Both regions are supplied with blood by a renal artery and drained by a renal vein. Within the cortex and medulla lie tightly packed excretory tubules and associated blood vessels. The excretory tubules carry and process a filtrate produced from the blood entering the kidney. Nearly all of the fluid in the filtrate is reabsorbed into the surrounding blood vessels and exits the kidney in the renal vein. The remaining fluid leaves the excretory tubules as urine, is collected in the inner **renal pelvis**, and exits the kidney via the ureter.

Weaving back and forth across the renal cortex and medulla are the **nephrons**, the functional units of the vertebrate kidney. Of the roughly 1 million nephrons in a human kidney, 85% are **cortical nephrons**, which reach only a short distance into the medulla. The remainder, the **juxtamedullary nephrons**, extend deep into the medulla. Juxtamedullary nephrons are essential for production of urine that is hyperosmotic to body fluids, a key adaptation for water conservation in mammals.

arranged in a highly organized manner and are closely associated with a network of capillaries. The vertebrate excretory system also includes ducts and other structures that carry urine from the tubules out of the kidney and, eventually, the body.

Vertebrate kidneys are typically nonsegmented. However, hagfishes, which are jawless vertebrates (see Concept 34.2), have kidneys with segmentally arranged excretory tubules. Because hagfishes and other vertebrates share a common chordate ancestor, it is possible that the excretory structures of vertebrate ancestors were also segmented.

We conclude this introduction to excretory systems with an exploration of the anatomy of the mammalian kidney and associated structures (Figure 44.12). Familiarizing yourself with the terms and diagrams in this figure will provide you

with a solid foundation for learning about filtrate processing in the kidney, the focus of the next section of this chapter.

CONCEPT CHECK 44.3

1. Compare and contrast the ways that metabolic waste products enter the excretory systems of flatworms, earthworms, and insects.
2. Where and how does filtrate originate in the vertebrate kidney, and by what two routes do the components of the filtrate exit the kidney?
3. **WHAT IF?** Kidney failure is often treated by hemodialysis, in which blood diverted out of the body is filtered and then allowed to flow on one side of a semipermeable membrane. Fluid called dialysate flows in the opposite direction on the other side of the membrane. In replacing the reabsorption and secretion of solutes in a functional kidney, the makeup of the starting dialysate is critical. What initial solute composition would work well?

For suggested answers, see Appendix A.